

Autonomous Research Agent

Labeled Test Traces & Failure Injection Analysis

Verification Scope: This document tracks execution traces across three failure and execution archetypes. **TRACE-001** verifies deterministic mock orchestration in the automated test suite; **TRACE-002** documents deliberate synthetic fault injection and retry recovery; **TRACE-003** documents an actual live production HTTP 404 failure and Gemini-driven dynamic replanning.

1 EXECUTION TRACE MATRIX

Trace ID	Environment	Trigger / Fault Condition	Expected Decision	Observed Decision	Final Outcome	Verification Source
TRACE-001 MOCK_NORMAL	Mock Isolated (Deterministic)	Nominal loop execution; 0 errors	CONTINUE → COMPLETE	CONTINUE → COMPLETE	Completed multi-step loop, state populated	tests/test_agent_loop.py::test_full_autonomous_loop_execution
TRACE-002 SYNTHETIC_TIMEOUT	Mock Fault Injection (CLI Flag)	Transient search timeout on call 1	RETRY (attempt 1/2)	RETRY → CONTINUE	Recovered on attempt 2, completed report	Captured CLI: python app/main.py --simulate-failure timeout
TRACE-003 LIVE_404_REPLAN	Live Production (Gemini + Tavily)	Live HTTP 404 on candidate URL	REPLAN	REPLAN → Revised plan	Replanned 4 search tasks, completed report	Captured live run (separate benchmark execution)

2 TRACE EXECUTIONS

TRACE-001 — MOCK_NORMAL (Automated Integration Test)		Deterministic Mock Test
Target: tests/test_agent_loop.py::test_full_autonomous_loop_execution	Providers: MockLLMProvider + MockSearchProvider	Assertions: is_complete == True, step_count ≥ 2, evidence > 0
[LangGraph Execution Flow: tests/test_agent_loop.py] 1. START → analyze_goal: MockLLMProvider generates deterministic GoalAnalysis 2. generate_plan: MockLLMProvider produces LLMPlan (Task 1: web_search Task 2: page_fetcher) 3. execute_step (Task 1): MockSearchProvider returns 3 synthetic SearchResultItem records [OBSERVATION] Retrieved 3 search results → [DECISION] CONTINUE (Advancing to step 2 of 2) 4. execute_step (Task 2): Mock page_fetcher returns 38-char FetchedPage [OBSERVATION] Extracted 38 chars → [DECISION] COMPLETE (All tasks executed) 5. process_evidence: Ingests chunks, filters by keyword density (≥0.30), deduplicates 6. synthesize_report → export_report: ResearchReport written to reports/ VERIFIED ASSERTIONS: assert final_state["is_complete"] is True and final_state["status"] == "completed"		
TRACE-002 — SYNTHETIC_TIMEOUT_RETRY (Deliberate Fault Injection)		Synthetic Fault Injection
Command: python app/main.py "Research agentic workflows" --provider mock --simulate-failure timeout	Mechanism: MockSearchProvider._simulated_timeouts_count == 0 emits timeout	
[STEP 1] Task: Search authoritative sources regarding Research agentic workflows Tool: web_search [web_search] [INFO] - [MockSearchProvider] Simulating induced transient timeout failure. ✗ Failed: Search failed: Request timed out after 15.0s (Simulated Failure for Assessment Demo). [DECISION] RETRY - Transient failure detected ('Request timed out after 15.0s...'); retrying 'task_1' (attempt 1/2). [STEP 2] (RETRY) Task: Search authoritative sources regarding Research agentic workflows Tool: web_search ✓ Completed: Retrieved 3 search results for 'Research agentic workflows latest developments' [DECISION] CONTINUE - Task completed successfully. Advancing to step 2 of 2. [STEP 3] Task: Extract detailed content from top discovered source Tool: page_fetcher ✓ Completed: Extracted 360 chars → [DECISION] COMPLETE (All tasks executed) RECOVERY OUTCOME: 1 transient failure intercepted, cleanly healed on retry, completed final synthesis.		

TRACE-003 — LIVE_404_REPLAN (Observed Live Failure & Autonomous Replanning)

Real Live Execution

Environment: Live Gemini 3.5 Flash-Lite + Tavily API Trigger: Candidate URL endpoint returned HTTP 404 Not Found Recovery: REPLAN decision triggered dynamic revised plan via Gemini (4 web_search tasks)

[STEP 1] web_search: 'agentic AI systems survey architectures' → ✓ Completed (5 results) → [DECISION] CONTINUE

[STEP 2] page_fetcher: Target 'https://example.com/research/1'

[page_fetcher] [WARNING] - Failed to fetch https://example.com/research/1: HTTP 404: Not Found

x Failed: Fetch failed for https://example.com/research/1: HTTP 404: Not Found

[DECISION] REPLAN - Task 'task_2' could not be completed (HTTP 404: Not Found). Triggering dynamic replanning.

REPLAN TRIGGERED

Replanning Active Strategy: Task 'task_2' failed with HTTP 404: Not Found. Triggering dynamic replanning via Gemini. Formulating 4 replacement tasks.

[planner] [INFO] - Formulating revised plan via Gemini 3.5 Flash-Lite...

REVISED PLAN ACTIVE: (4 Tasks: 1. Broader surveys | 2. Planning mechanisms | 3. Tool use | 4. Memory & reflection)

[STEP 3] web_search: 'recent advances agentic AI systems survey arxiv' → ✓ 5 results → [DECISION] CONTINUE

[STEP 4] web_search: 'agentic AI planning reasoning tree of thought' → ✓ 5 results → [DECISION] CONTINUE

[STEP 5] web_search: 'agentic AI tool use dynamic API function calling' → ✓ 5 results → [DECISION] CONTINUE

[STEP 6] web_search: 'agentic AI memory architectures self correction' → ✓ 5 results → [DECISION] COMPLETE

► EVIDENCE PROCESSING: Ingested 52 candidates → 38 relevant → 1 duplicate removed → 37 final evidence items.

► REPORT SYNTHESIS: Synthesized structured report across 21 cited sources. Status: COMPLETED.

3 MOCK INFRASTRUCTURE, FAULT INJECTION & TEST ASSERTIONS

Mock Generation & Fault Injection Mechanics:

- **MockLLMProvider:** Intercepts generate_structured(), returning typed deterministic Pydantic schemas (GoalAnalysis, LLMPlan, ResearchReport). Requires zero external API keys.
- **MockSearchProvider:** Simulates realistic search responses without network I/O. Supports --simulate-failure timeout, empty, and error.
- **Transient Failure Counter:** MockSearchProvider._simulated_timeouts_count tracks retry attempts, failing call 1 and passing call 2 to verify autonomous self-healing.
- **Trace Labeling:** Traces are uniquely labeled by environment and scenario (MOCK_NORMAL, SYNTHETIC_TIMEOUT_RETRY, LIVE_404_REPLAN) with deterministic state tracking.

Verified Automated Test Assertions:

- **SSRF Guardrails (test_tools.py):** Blocks RFC 1918 private IPs (10.0.0.1, 192.168.1.1), loopback (127.0.0.1, ::1), and cloud metadata endpoints (169.254.169.254).
- **Evaluator Matrix (test_agent_loop.py):** Validates that transient errors trigger RETRY up to 2 times, while 404 or exhausted retries trigger REPLAN, and step ≥ 10 triggers FAIL.
- **Closed Schemas (test_planning_schema.py):** Asserts LLMPlan has no additionalProperties, ensuring strict Gemini Developer API compliance.
- **Evidence Quality (test_evidence.py):** Asserts keyword density filtering (≥ 0.30) and SHA-256 + token Jaccard deduplication (≥ 0.75).